

VENKATA PUVVALA

Jersey City, New Jersey | (+1) 551-359-3551 | hemanthp1103@gmail.com | [LinkedIn](#)

PROFESSIONAL SUMMARY

- **Senior Data Engineer** with over **10 years** experience in **IT**, specializing in **Big Data, ETL Development, Data Modeling, and Cloud Data Engineering** across industries such as Finance, Healthcare, and Insurance.
- Expert in building cloud-native **ETL pipelines** on **GCP (BigQuery, Dataflow), AWS (Redshift, Glue), and Azure (Data Factory, Synapse)**.
- Hands-on experience in **Real-time streaming (Kafka, Spark Streaming)** and **workflow automation (Apache Airflow, Cloud Composer)**.
- Reduced **AWS S3-based storage costs by 20%** and improved **BigQuery query speeds by 40%**.
- Enhanced storage efficiency by configuring GCP's Cloud Storage buckets and object lifecycle policies, resulting in reduced storage costs while maintaining data availability and durability.
- Led migration of on-premise **Hadoop** workflows to Google Cloud (BigQuery, DataProc) and AWS (Redshift, Glue).
- Designed and implemented scalable data lake architectures using **Delta Lake, Iceberg, and Apache Hudi**, improving data governance, versioning, and ACID compliance for large-scale analytics.
- Led the development of **CI/CD pipelines** for automated deployment of data workflows using **Terraform, Kubernetes, and Jenkins**, reducing deployment time by 40% and improving operational efficiency.
- Skilled in optimizing Spark jobs and BigQuery queries, ensuring high performance and cost-effective cloud operations.
- Expertise in database development and data modeling, with a strong understanding of **Star Schema, Snowflake Schema, and Fact & Dimension tables**.
- Strong background in **BI & Data Visualization**, with expertise in **Power BI, Tableau, Looker, and Data Studio**, helping businesses make informed, data-driven decisions.
- Extensive experience working with **NoSQL databases**, including **MongoDB, Cassandra, and Azure Cosmos DB**, for flexible and scalable data storage solutions.
- Comfortable with **Linux shell scripting, RDBMS development (SQL Server, PostgreSQL, Oracle)**.
- Proficient in **Python (Pandas, NumPy, Scipy, PySpark), SQL (T-SQL, PL/SQL, PostgreSQL), and Scala**, with a deep understanding of data engineering best practices.
- Experienced in migrating and transforming large datasets, seamlessly transitioning from on-premises ETLs to cloud-native architectures with minimal downtime.
- Skilled in batch data processing, leveraging **Hadoop, MapReduce, YARN, Pig, and Hive** to handle high-volume data workloads efficiently.
- Passionate about cloud cost optimization and monitoring, utilizing tools like **GCP StackDriver, AWS CloudWatch, and Azure Monitor** to ensure efficient resource usage.
- Have Extensive Experience in IT data analytics projects and hands-on experience in migrating on-premises ETLs to **Google Cloud Platform (GCP)** using cloud-native tools such as **BIG query, Cloud DataProc, Google Cloud Storage, and Composer**
- Experienced in building and managing data catalogs, ensuring proper data lineage, metadata management, and enterprise-wide accessibility.
- Proficient in designing, implementing, and managing **Infrastructure as Code (IaC)** using **Terraform and CloudFormation**, enabling scalable, automated, and reproducible cloud deployments.
- Proven ability to **mentor junior engineers and lead data engineering teams**, fostering a culture of continuous learning and innovation.

TECHNICAL SKILLS

Programming Languages	Python (Pandas, NumPy, SciPy, PySpark), SQL (T-SQL, PL/SQL, PostgreSQL), Scala,
Big Data & Cloud Platforms	GCP (BigQuery, DataProc, Cloud Composer), AWS (Redshift, Glue, S3), Azure (Data Factory, Synapse)Hadoop (HDFS, YARN, MapReduce, Hive, Pig, Sqoop, HBase, Cloudera, Hortonworks)
Workflow Orchestration	Apache Airflow, GCP Cloud Composer, Oozie
Relational Databases & Data Modeling	SQL Server, PostgreSQL, Oracle, Star Schema, Snowflake Schema, Fact & Dimension Tables
NoSQL Databases	MongoDB, Cassandra, Azure Cosmos DB
Streaming & Real-time Data Processing	Apache Kafka, Spark Streaming, Google Cloud Pub/Sub, Flink
Processing Frameworks	Apache Spark, PySpark, Databricks
DevOps & CI/CD for Data Pipelines	Terraform, GitHub Actions, Jenkins, Docker, Kubernetes, AWS CloudFormation
BI & Data Visualization	Tableau, Power BI, Looker, Google Data Studio
Operating Systems	Windows, Linux, UNIX (HP-UX, Solaris, AIX)

PROFESSIONAL EXPERIENCE

Senior Data Engineer

January 2022 - Current

Amazon, Arlington, VA (Remote)

- Led all phases of **Data Mining**, including **Data Collection, AWS-Based Data Cleaning, Model Development, Validation, and Visualization**, while performing **Gap Analysis**.
- Optimized **SQL queries** and database partitioning strategies in **BigQuery, Redshift, and Snowflake**, reducing query latency by 40%.
- Created and optimized **AWS Lambda functions** and **API Gateways**, enabling seamless data submission and interaction between **serverless applications**.
- Implemented row-level security and access control in **BigQuery & Redshift** to ensure data governance.
- Conducted **data mapping** between source and target systems, developing **logical data models, ER diagrams, and class diagrams**, and optimized **SQL queries** for data transformation.
- Processed **Avro files** using **Apache Kafka** and performed **advanced analytics with Spark and Scala**.
- Developed feature engineering and data pipelines for ML models, integrating structured/unstructured data into **Vertex AI** and **SageMaker**.
- Reverse-engineered **backend databases** to optimize **T-SQL scripts**, enhancing performance by refining **Views, Stored Procedures, Triggers, and Functions**.
- Developed and optimized ETL pipelines to ingest data into cloud-based data warehouses, supporting BI teams using **Power BI, Tableau, and Looker**.

- Designed data marts and OLAP structures in BigQuery & Redshift to support analytics workloads.
- Developed **Python automation scripts** to handle **Data Subject Requests (DSRs)** from **AWS Snowflake tables**, integrating them with **Adobe Analytics Privacy API** for compliance tracking.
- Designed a **Serverless Architecture (AWS Lambda, API Gateway, S3)** that scaled to process 1M+ data records daily.
- Managed data ingestion and transformation processes in **Amazon Redshift**, using **AWS CloudWatch to monitor AWS RDS instances** for performance optimization.
- Reduced AWS storage costs by 20% and improved query speeds by 40% using **Apache Hudi** for optimized **S3-based data lakes**.
- Extracted **real-time streaming data** using **Kafka and Spark Streaming**, transforming **D-streams into RDDs** and storing structured data in **Cassandra** for high availability.
- Designed and deployed **ETL pipelines** to extract, transform, and load data from **SAP S4 into SAP BusinessObjects**, enabling **advanced analytics and reporting**.

Technologies: AWS (Lambda, Redshift, Kinesis, Snowflake, S3, Glue), GCP (BigQuery, Cloud Storage, Pub/Sub), Apache Kafka, Spark, Apache Airflow, SQL Server 2022, Oracle 12c, Python, Tableau, Power BI

Senior Data Engineer

June 2019 - December 2021

Walmart, Sunnyvale, CA

- Designed and implemented scalable **data pipelines and ETL processes** using **Python, SQL, Apache Spark, and Hadoop**, ensuring efficient data transformation and integration to support business intelligence and analytics.
- Developed data models and analytics frameworks in **GCP**, improving data storage and retrieval strategies for business intelligence applications.
- Leveraged **BigQuery, Dataflow, and Airflow (GCP Composer)** to architect and deploy **cloud-based data workflows**, ensuring high availability, performance, and cost efficiency.
- Developed **data models, data marts, and relational structures** for analytics and reporting, optimizing data storage and retrieval strategies.
- Collaborated with **data scientists, analysts, and business stakeholders** to **identify data sources, define data strategies, and develop solutions** that aligned with business objectives.
- Built high-performance **Hive SQL scripts for database modeling**, optimizing **partitioning, clustering, and indexing** to improve query performance and data warehouse efficiency.
- Designed and implemented **Big Data solutions** using **Apache Spark (PySpark, Scala), Hive, and Databricks**, reducing execution times and optimizing cloud computing costs.
- Developed and maintained **APIs for data extraction and transformation**, ensuring seamless integration across multiple platforms and services.
- Led **PoC implementations for Machine Learning models**, leveraging **Cloud ML and BigQuery ML** for predictive analytics and data quality enhancements.
- Implemented **data governance and quality frameworks** by monitoring and optimizing **Google Data Catalog, BigQuery billing analysis, and cloud cost management**.
- Developed **incremental data ingestion pipelines** using **Sqoop, Kafka Connect, and Apache Airflow**, facilitating smooth data migration between relational databases and Hadoop ecosystems.
- Designed and managed **Hadoop clusters in Cloudera distributions**, utilizing **Apache Spark, Hive, Pig, and Sqoop** for large-scale data processing.

- Led the **design, testing, and deployment of database solutions**, including **Stored Procedures, Views, and Triggers**, to support analytical workloads and enterprise reporting.
- Built **geospatial visualizations** and dashboards using **Python (GeoPandas, Folium), GCP tools (Looker, Data Studio)**, delivering actionable insights to stakeholders.
- Demonstrated expertise in **Kubernetes, Terraform, and cloud automation**, streamlining deployments and ensuring infrastructure reliability.
- Spearheaded **optimization of Databricks workflows and Spark processes**, significantly improving data processing efficiency while reducing operational costs.
- Developed and executed **enterprise-wide data governance policies**, ensuring data consistency, compliance, and security.
- Provided technical mentorship and guidance to **junior engineers and cross-functional teams**, fostering a culture of innovation and best practices in data engineering.

Technologies: GCP (BigQuery, Cloud Storage, Dataflow, Airflow, Looker), Python (GeoPandas, Pandas, Folium), SQL, Kafka Connect, Apache Airflow, Terraform, Apache Spark, Hive, Databricks, Sqoop, Hadoop (Cloudera), Looker, Data Studio, BigQuery ML

Senior Data Engineer - ETL
Wells Fargo, Charlotte, NC

July 2017 - May 2019

- Migrated an **entire Oracle database to BigQuery**, ensuring **optimized query performance and scalability**, and developed **Power BI reports for advanced data visualization**.
- Built and automated **ETL pipelines in Apache Airflow (GCP Composer)** for **data transformation, extraction, and loading (ETL) processes**, utilizing various **Airflow operators**.
- Hands-on experience in **GCP services**, including **DataProc, Google Cloud Storage (GCS), Cloud Functions, and BigQuery**, ensuring efficient **cloud data processing and storage**.
- Collaborated with the team to **develop a framework for generating daily ad-hoc reports and extracting enterprise data from BigQuery**, improving **business reporting capabilities**.
- Developed and maintained **scalable geospatial data solutions**, leveraging **Google Cloud Platform (GCP), Spark, ANSI SQL, and Python**, enabling **advanced geospatial analytics**.
- Engineered and optimized **geospatial data pipelines using PySpark**, enhancing **data processing efficiency** for large-scale geospatial datasets.
- Implemented **cloud-based geospatial data storage and processing solutions** to ensure **high availability, scalability, and performance** in GCP environments.
- Designed and executed **advanced geospatial queries using ANSI SQL** for **data extraction, transformation, and loading (ETL) processes**, supporting **analytical and business intelligence applications**.
- Automated **geospatial data workflows using Python scripts**, improving **data accuracy, consistency, and timeliness** for data-driven decision-making.
- Collaborated with **cross-functional teams** to integrate **geospatial data** into broader business operations, enhancing **decision-making with location-based insights**.
- Designed and coordinated with the **Data Science team** to implement **advanced analytical models in Hadoop clusters**, ensuring **efficient processing of large datasets**.
- Created **BigQuery authorized views** to enforce **row-level security**, restricting **data access based on user roles** and ensuring **compliance with data governance policies**.
- Skilled in **creating scalable and reliable Airflow environments**, including **cluster management, task scheduling, and workflow automation**.

- Strong understanding of **Airflow's security features**, including **LDAP authentication, OAuth integration, and SSL certificates**, ensuring **secure workflow execution**.
- Experienced in **troubleshooting and resolving Airflow-related issues**, including **task failures, DAG misconfigurations, and scalability challenges**.
- Developed and optimized **GCP-based data pipelines for healthcare data**, maintaining **data quality, security, and compliance with industry regulations**.
- Utilized **Databricks and PySpark for batch and streaming data processing**, delivering **real-time insights and enabling business intelligence solutions**.

Technologies: GCP (Cloud Storage, BigQuery, Composer, DataProc, Cloud SQL, Cloud Functions, Pub/Sub), Big Data (Spark, Azure Storage, Azure Database, Azure Data Factory, Azure Analysis Services), Python 3.0, Tableau, Git

Data Engineer

May 2012 - June 2015

VASR Infotech, India

- Designed and implemented **data pipelines using Apache Spark and Hive** to ingest **customer behavioral data into the Hadoop platform**, enabling **advanced user behavior analytics**.
- Developed **Spark applications in Scala** for **data ingestion between environments**, incorporating **test cases to validate data integrity and transformations**.
- Created and optimized **Hive tables** to efficiently load and transform **large datasets**, ensuring **seamless data processing and retrieval**.
- Automated and streamlined **data pipelines** to transfer **over 25 GB of data from Oracle to Hadoop and Google BigQuery**, utilizing **GitHub for source control and Jenkins for CI/CD**.
- Developed a **Python-based automation script** to detect and upload XML files from a **Linux directory to Google Cloud Storage**, ensuring **automated ingestion and seamless data processing in BigQuery**.
- Utilized **Google BigQuery SQL and Amazon Athena** to design and develop **reporting frameworks**, enabling **optimized data analysis and business intelligence solutions**.
- Built and deployed **Google Dataflow pipelines** to efficiently upload **large-scale public datasets** into **BigQuery**, ensuring **scalability and performance**.
- Implemented **end-to-end testing frameworks** between **Google Dataflow and BigQuery**, validating **data integrity and consistency**.
- Optimized **Hive tables** using **advanced performance enhancement techniques** such as **partitioning and bucketing**, significantly **reducing query execution times**.
- Provided expertise in **enterprise architecture strategies**, guiding **classification, implementation, growth, adoption, and compliance processes**.
- Designed and developed **highly scalable and available distributed systems**, ensuring **fault tolerance and high-performance data processing**.
- Worked extensively with **AWS services**, including **EC2, Lambda, SES, SNS, VPC, CloudFront, and CloudFormation**, to design and deploy **cloud-based Hadoop applications**.
- Created **detailed architecture blueprints and technical documentation**, defining **bill of materials (BoM)**, required **AWS services (EC2, S3, RDS)**, and **cloud-based data processing tools**.
- Developed and deployed **custom Hadoop applications** in an **AWS environment**, integrating **big data processing frameworks**.

- Hands-on experience with **AWS EC2, ECS, ELB, EBS, S3, VPC, IAM, SQS, RDS, Lambda, CloudWatch, Storage Gateway, CloudFormation, Elastic Beanstalk, and Auto Scaling**, ensuring **optimal performance and high availability**.
- Designed and implemented **Hive External Tables** for **semantic data storage**, allowing seamless **querying and data retrieval** using **Hive Query Language (HQL)**.
- Conducted **source-to-target data mapping**, performing **storage estimation** and providing **support for setup and partitioning strategies** to enhance **data efficiency**.
- Developed **Atomic workflows** to cleanse and transform **raw datasets** into **structured information**, optimizing **HDFS-based storage and processing**.
- Utilized **HDFS File System API** to integrate with **FTP servers**, leveraging **AWS S3 SDK** for efficient **cloud-based data storage and retrieval**.
- Configured and deployed **multi-node Hadoop clusters** on **Amazon EC2**, ensuring **optimal performance and scalability** for **big data processing**.
- Deployed and managed **HBase clusters** in **AWS**, configuring **scalable nodes** based on **business requirements** to **improve NoSQL storage performance**.
- Led the development of **distributed data solutions** using **MapR Hadoop distribution**, ensuring **high-performance parallel data processing**.

Technologies: AWS (EC2, S3, Lambda, RDS, CloudFormation, CloudWatch), Hadoop (HDFS, HBase, Hive, Spark, MapReduce), Google Cloud (BigQuery, Dataflow, Cloud Storage), SQL, Python, Jenkins, GitHub